

Bank of America Content Fragment Merge

Service User:

bofa-sync-service

- service user created via RepositoryInitializer
(ui.config/src/main/content/jcr_root/apps/bofa-sync/osgiconfig/config.author/org.apache.sling.jcr.repoinit.RepositoryInitializer~bofa-sync-users.cfg.json)
- Permissions set via ACL currently not working as expected
- Needs the following permissions:
 - o read, write: /content/dam
 - o read, write: /var/bofa-sync
 - o read: /conf/bofa-sync
 - o The service user needs read access to the Content Fragment model location (demo code is located in /conf/bofa-sync)

Merge data:

- stored in /var/bofa-sync/merge
- Temporarily location for incoming Content Fragments: /var/bofa-sync/merge/tmp
- Paths are created via RepositoryInitializer
(ui.config/src/main/content/jcr_root/apps/bofa-sync/osgiconfig/config.author/org.apache.sling.jcr.repoinit.RepositoryInitializer~bofa-sync-users.cfg.json)

UI:

- Content Fragment Merge
 - o <http://localhost:4502/content/bofa-sync/en/cf-list-merge.html>
 - o Lists all Content Fragments and Variations that need to be merged. This page shows the status of each merge (open/closed), has a link to the Content Fragment editor, a link to the diff page, and the ability to auto merge if available
- Content Fragment Diff

- <http://localhost:4502/content/bofa-sync/en/cf-diff.html>
 - ex: <http://localhost:4502/content/bofa-sync/en/cf-diff.html?contentFragmentPath=/content/dam/bofa/content-fragments/samples/car-sample&variationName=master>
 - contentFragmentPath: JCR path of Content Fragment
 - variationName: Content Fragment variation name
- Shows all the fields for the Source and Destination Content Fragment, and the diff of all the fields. It also shows the merge status, a link to the editor, and ability to close the merge

HTTP Rest Endpoints:

- [/bin/packageReader](#)
 - ex:
 http://localhost:4502/bin/packageReader?packagePath=/etc/packages/my_packages/sample.zip
 - packagePath: JCR location of content package containing incoming changes
 - This will:
 - Read the content package, it currently prepends the temp path (/var/bofa-sync/merge/tmp) to the Assets in the package
 - ex: a CF in /content/dam/bofa/content-fragments/samples/car-sample will be stored in /var/bofa-sync/merge/tmp/content/dam/bofa/content-fragments/samples/car-sample
 - Start the Merge Workflow, set the status to all CF Variations to "open"
 - Servlet:
 [core/src/main/java/com/adobe/aem/acs/bofa/core/servlets/ContentPackageReader.java](#)

- [/bin/autoMerge](#)
 - ex:
 <http://localhost:4502/bin/autoMerge?contentFragmentPath=/content/dam/bofa/content-fragments/samples/car-sample&variation=master>
 - contentFragmentPath: JCR path of Content Fragment
 - variation: Content Fragment variation name
 - This will:

- Check to make sure the Content Fragment is eligible for auto merge (all fields contain either all additions or deletions)
 - Any fields that have changed, take the Source (incoming) content and put it in the Destination (current system /content/dam) fields
 - Servlet:
core/src/main/java/com/adobe/aem/acs/bofa/core/servlets/AutoMergeServlet.java
- /bin/closeContentFragmentMerge
 - ex:
http://localhost:4502/bin/closeContentFragmentMerge?contentFragmentPath=/content/dam/bofa/content-fragments/samples/car-sample&variationName=master
 - contentFragmentPath: JCR path of Content Fragment
 - variation: Content Fragment variation name
 - This will:
 - Check to make sure the Content Fragment Source and Destination content is the same
 - Update the merge status to "closed"
 - ex: /var/bofa-sync/merge/tmp/content/dam/bofa/content-fragments/samples/car-sample/jcr:content/data/master (mergeStatus: closed)
 - Servlet:
core/src/main/java/com/adobe/aem/acs/bofa/core/servlets/CloseContentMergeServlet.java
- /bin/endMerge
 - ex: http://localhost:4502/bin/endMerge
 - This will:
 - Update the merge status to "closed"
 - ex: /var/bofa-sync/merge/jcr:content (status: closed)
 - Remove all temporary Content Fragment nodes under /var/bofa-sync/merge/tmp
 - Servlet:
core/src/main/java/com/adobe/aem/acs/bofa/core/servlets/EndMergeServlet.java

Java Services

- core/src/main/java/com/adobe/aem/acs/bofa/core/services/impl
- ContentFragmentServiceImpl.java
 - getContentFragmentInfo()
 - returns ContentFragmentInfo for the given contentFragmentPath
 - getContentFragmentInfoList()
 - returns List<ContentFragmentInfo> for all Content Fragments that will be merged
 - getContentFragmentsInTempPath()
 - Finds all Content Fragments to be merged in the temp path, this is only called once when a new merge workflow starts
- ContentFragmentMergeWorkflowServiceImpl.java
 - getContentFragmentMergeStatus()
 - returns the status of the merge workflow: open/closed
 - isContentFragmentMergeInProgress()
 - returns true if there is an open merge
 - startContentFragmentMerge()
 - this is called when a new merge workflow is started, it finds all Content Fragments to be merged and sets the merge status on all variations
 - closeContentFragmentMerge()
 - sets the status to "closed" for the given variation
 - endContentFragmentMerge()
 - closes the merge workflow, deletes all temporary Content Fragments
- ContentFragmentMergeServiceImpl.java
 - getContentFragmentMerge()
 - return ContentFragmentMerge for the given contentFragmentInfo
 - getContentFragmentMergeList()
 - returns List<ContentFragmentMerge> for the given list of contentFragmentInfoList
 - autoMerge()
 - checks to see if Source and Destination Content Fragments to can auto merged and updates the corresponding values
- JavaDiffUtilDiffServiceImpl.java
 - diff()

- performs diff of 2 fields

Metadata Audits in TimeLine

1. Listeneres location ::
com.adobe.aem.acs.bofa.core.listeners.MetadataUpdateListener.java
Change it to appropriate location based on BofA package.
2. Change system user from 'datawrite' to appropriate BofA System user.
3. System user should have read + write access to below paths
/content/dam
/var/audit/com.day.cq.dam
4. Go to Felix Console configurations.
<http://localhost:5502/system/console/configMgr>

- Search for “Apache sling Service User Mapper Service Amendment” configuration.
- Click on Plus sign against "Apache sling Service User Mapper Service Amendment" to create new factory configuration.
- Enter Service Mapping Details
Enter entry in form of BundleId:subserviceName=userName.

Lets Assume below details:-

bundleId = aem-guides-wknd.core (You can find your bundle symbolic name under your core pom.xml)

subServiceName = datawrite (Used in your java for getting resource resolver based on subService Name)

systemUser Name = datawrite (Your system user Name, that you have given at time of creation of system user)

so your Service Mapping entry will look like this aem-guides-wknd.core=[datawrite]

The screenshot shows the configuration page for the 'Apache Sling Service User Mapper Service Amendment'. It includes a 'Ranking' field set to 0, a 'Service Mappings' table with one entry 'aem-guides-wknd.core=[datawrite]', and a 'Configuration Information' section with fields for Persistent Identity (PID), Factory Persistent Identifier (Factory PID), and Configuration Binding.

Configuration Information	
Persistent Identity (PID)	org.apache.sling.serviceusermapping.impl.ServiceUserMapperImpl.amended.f14efa0f-f6c0-4e47-abd8-fc5b98487427
Factory Persistent Identifier (Factory PID)	org.apache.sling.serviceusermapping.impl.ServiceUserMapperImpl.amended
Configuration Binding	Unbound or new configuration

5. After deploying code and once bundle is active -
Go to Felix Console components

3835	▼ com.adobe.aem.guides.wknd.core.listeners.MetadataUpdateListener	active
	Bundle	aem-guides-wknd.core (838)
	Implementation Class	com.adobe.aem.guides.wknd.core.listeners.MetadataUpdateListener
	Default State	enabled
	Activation	immediate
	Configuration Policy	optional
	serviceld	7917
	Service Type	singleton
	Services	javax.jcr.observation.EventListener
	PID	com.adobe.aem.guides.wknd.core.listeners.MetadataUpdateListener
	Reference repository	Satisfied
		Service Name: org.apache.sling.jcr.api.SlingRepository
		Cardinality: 1..1
		Policy: static
		Policy Option: reluctant
		Bound Service ID: 1355 (com.adobe.granite.repository.impl.SlingRepositoryManager)
	Properties	component.id = 3835
		component.name = com.adobe.aem.guides.wknd.core.listeners.MetadataUpdateListener
		event.topics = org.apache.sling.api.resource.Resource/CHANGED
		event.types = *
		service.description = Metadata Update Listener

<http://localhost:5502/system/console/components> and search for 'MetadataUpdateListener' you will see Listener active

6. Go to SlingLog to configure debug log for MetadataUpdate

<http://localhost:5502/system/console/slinglog>

For debugging create metadata-audit.log configured to

com.adobe.aem.acs.bofa.core.listeners.MetadataUpdateListener